COMP 441 | LABORATORY REPORT

Lab 8: Visual Regression & Accessibility Testing

Screenshot comparison, WCAG scan, and remediation results

Target: Luxe Shop demo (user-provided hosted sample)
Prepared: 25 October 2026 **Tools:** Playwright 1.64.1 / Chromium, axe-core 4.14.0, pixelmatch 6.0.0

Result: baseline scans found 28 serious contrast-failure instances across light and dark states. Local CSS overrides cleared every axe violation in five tested page/theme states. The deliberate heading-spacing change produced measurable desktop and mobile pixel diffs.

Scope and method

Axe scanned Home, Shop, About, and Cart in light mode, plus Shop in dark mode. The scan included WCAG 2.0/2.1 Levels A and AA rules. The visual test compared the Shop page before and after a local CSS change at desktop (1280 x 900) and mobile (390 x 844), using pixelmatch threshold 0.1.

Baseline accessibility results

Page / theme	Axe passes	Violations	Affected nodes	Finding
index.html / light	24	1	4	color-contrast (serious)
shop.html / light	26	1	12	color-contrast (serious)
about.html / light	25	0	0	None
cart.html / light	25	0	0	None
shop.html / dark	26	1	12	color-contrast (serious)

Triage table ? two distinct accessibility defects

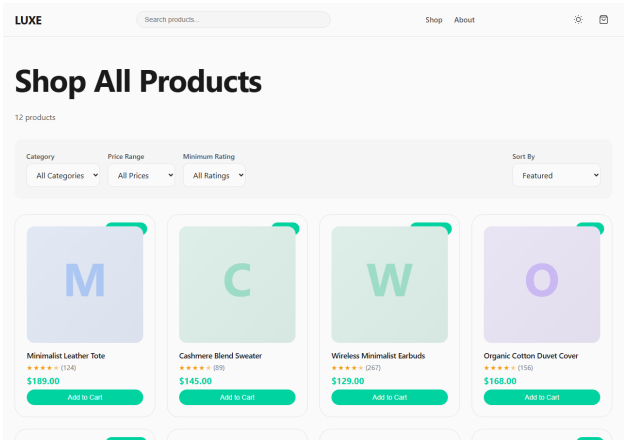
Affected elements / state	WCAG criterion / level	Impact	Measured contrast	Remediation
Product prices, Home + Shop / light (16 nodes)	2.1 SC 1.4.3 AA	Serious	#00D3A0 on #FAFAFA 1.85:1; minimum 4.5:1	Set price text to #006B50. New contrast: 6.25:1.
Review-count text, Shop / dark (12 nodes)	2.1 SC 1.4.3 AA	Serious	#737373 on #2E2E2E 2.86:1; minimum 4.5:1	Set review-count text to #BDBDBD. New contrast: 7.23:1.

The raw before/after axe results are included as JSON. Axe reported one incomplete check item per scan state; those are not counted as violations and remain candidates for manual review.

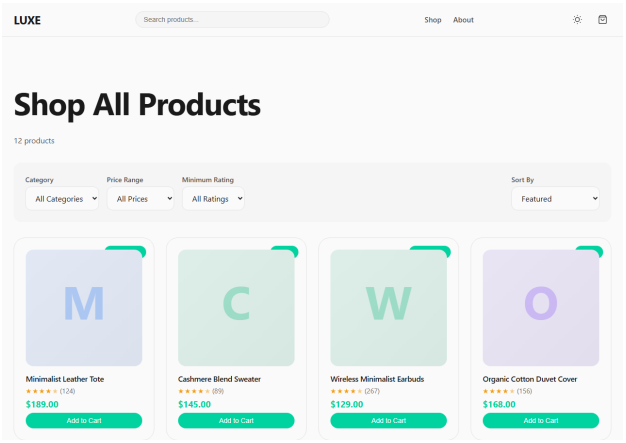
Visual regression experiment

A deliberate spacing change was applied only in the test browser: `main h1 { margin-top: 48px !important; }`. The live hosted site was not changed. Screenshots use the same viewport dimensions before and after.

Desktop scenario ? Shop, 1280 x 900

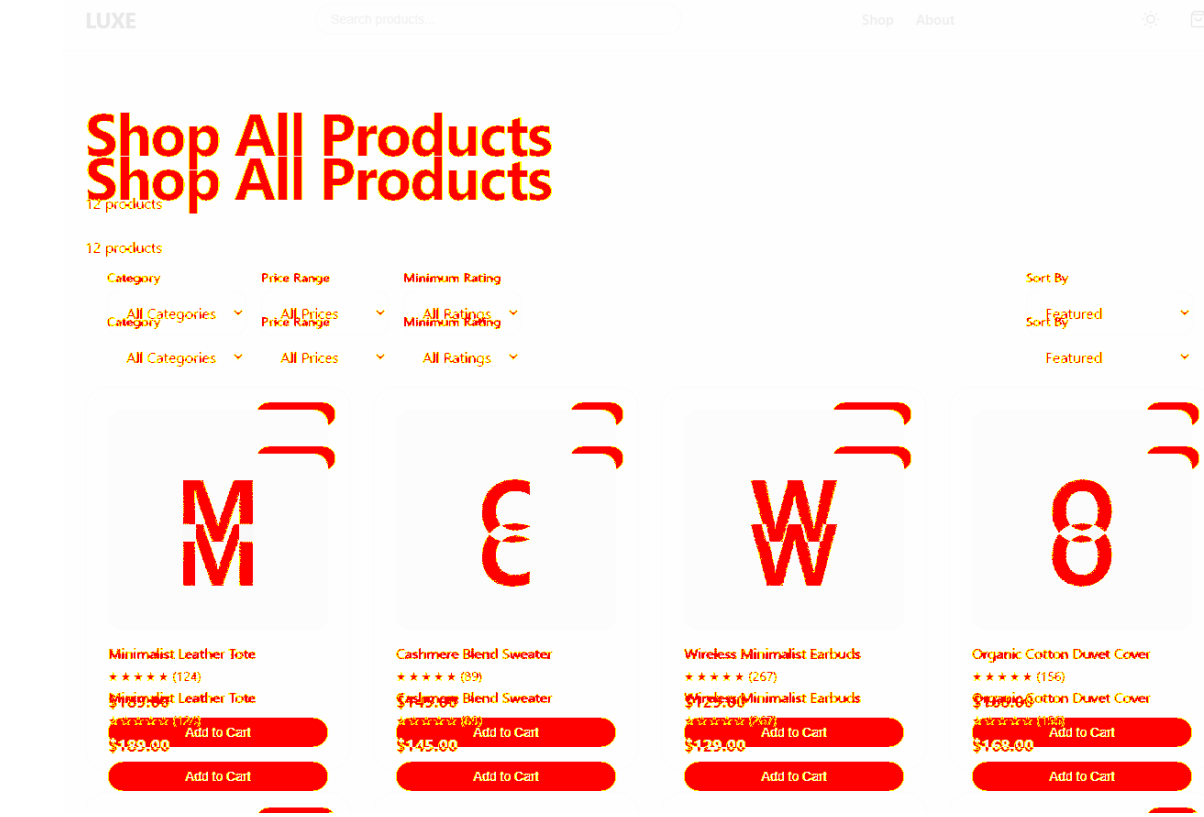


Reference / before



Deliberate CSS change / after

Pixel difference (red marks changed pixels)



Scenario	Viewport	Changed pixels	Changed area
desktop	1280 x 900	106,075 / 1,152,000	9.2079%
mobile	390 x 844	18,697 / 329,160	5.6802%

The mobile baseline, changed screenshot, and diff are also included in this folder. Full pixel metrics are in `visual-diff-report.json`.

Accessibility remediation and verification

The test harness injects the scoped rules in `accessibility-fixes.css` after the baseline scan, then repeats the same axe checks. The product-price rule covers both light-mode pages; the review-count rule applies only in dark mode.

Page / theme	Passes	Violations	Affected nodes
index.html / light	24	0	0
shop.html / light	26	0	0
about.html / light	25	0	0
cart.html / light	25	0	0
shop.html / dark	26	0	0

Fix summary

Fix 1 ? light-mode product prices: replaced the bright mint text with dark green (#006B50), raising contrast from 1.85:1 to 6.25:1 on the light product-card background. Applied to all 16 flagged price elements on Home and Shop.

Fix 2 ? dark-mode review counts: changed muted gray text to #BDBDBD, raising contrast from 2.86:1 to 7.23:1 on the dark cards. Applied to all 12 flagged review-count elements on Shop.

Verification: after the overrides, the five page/theme scans recorded zero axe violations and zero affected nodes. The final results are in `axe-after.json`.

Reflection

A pixel-diff test cannot determine whether an image has meaningful alternative text, a form control has a programmatic label, a control is keyboard-operable, focus order is logical, or a component exposes the right role and state. It only reports rendered pixel changes, not whether the interface is perceivable or operable. Contrast violations are visually present, but pixel diffs alone do not calculate WCAG ratios or identify which users are affected. Therefore visual comparison and semantic/accessibility testing complement rather than replace one another.

Scope note: the target is a third-party hosted demo. The CSS remediations are browser-test overrides, not a deployed source-code change. The reproducible harness is `lab8.mjs`; run `npm install`, `npx playwright install chromium`, then `npm test` from this deliverables folder.